From Trajectories to Differentiated Skills: Cognitive Differentiation for Self-Improving Agents

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HarnessRSI 64 ★★

evidence: retained

Paper

1 What changes, what is kept

1. Evaluated training trajectories
Filter runtime noise; keep trace evidence.
Success: preserve. Failure: supported recovery.

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2. Retain and assign semantic gradients
Drop generic, unsupported or sample-only findings.
One primary role; at most one secondary role.
Each update keeps its supporting trace lines.

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3. Maintain four functional components
Meta: task-wide control and stopping rules
Experience: local heuristics and recovery
Tool: capability-specific operational guidance
Script: deterministic procedures + I/O contracts

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4. Validate, coordinate, then publish
Scripts: self-test + independent verification.
Failed edit-contract checks: no publication.
Not a task-score-improvement acceptance gate.

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5. Agent selects relevant components
Descriptions visible before loading guide selection.
Weights and improvement protocol stay fixed.

2 Highest scores in all 36 settings

+7.6 pp vs Base · +2.6 pp vs best rival

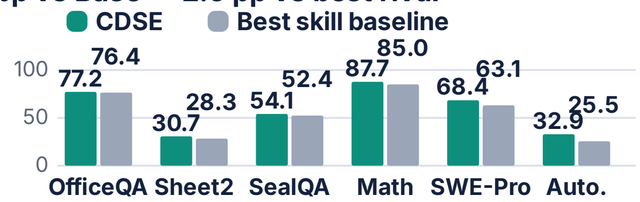


Chart: GLM-5.2 / Claude Code test scores, Table 1, after three evolution rounds. Rivals: SkillOpt-lite, Trace2Skill or EvoSkill (best per column).

Headline gains macro-average within-benchmark changes over models and harnesses; no variance, CIs or run/seed counts reported.

4 Where gains are small or fail

- **Thin margins:** MiniMax-M3/Codex Sheet2 29.4 vs SkillOpt-lite 29.1; GLM/Claude OfficeQA 77.2 vs 76.4. No uncertainty to assess significance.
- **Uneven gain:** Automation vs Base: Claude Code +12.1/+14.6/+10.9 pp; Codex +4.9/+8.4/+5.5 (GLM/DeepSeek/MiniMax).
- **Later rounds can regress:** GLM/Claude SWE 69.0 → 68.4 (R2→R3); same corpus reused, not fresh experience.
- **Transfer failure:** DeepSeek SealQA 48.5 transferred vs 51.8 Base (59.4 native). Spreadsheet Script → SWE: −3.9 pp.

3 Controls that matter

GLM-5.2 / Claude Code · mechanism ablations

Table 3 · score (%)	Sheet2	SWE	Auto.
Full CDSE	**30.7**	**68.4**	**32.9**
No differentiation	25.8	62.6	24.3
Shared updater	27.1	65.4	28.0
No coordination	29.6	67.5	30.7
No adaptive interface	26.4	64.3	27.1

No-differentiation uses the same semantic gradients. Table 3 reports no uncertainty or run counts.

- **Split:** training:test = 2:7 within each selected benchmark; training traces only. Same task families; no group-disjoint split or overlap audit reported.
- **Baseline access:** same training split, instances, tools and context limit. Trace2Skill merges success + error; EvoSkill: error-only, 1.5 epochs, frontier 3; SkillOpt-lite: 10 rounds, batch 10. CDSE library/gates not stated as shared.
- **Tuning:** original baseline defaults; no default-selection split or tuning budgets stated. Capacity values, runtime limits and evolution decoding settings omitted.
- **Cost (one round only):** CDSE input/output tokens (M): Sheet2 50.90/2.800; SWE 127.80/5.800; Auto. 24.70/1.100. Similar baseline totals; not exact compute matching.

Takeaway

Functional organization beats a unified skill even with the same extracted lessons. But transfer is selective, later fixed-corpus rounds can regress, and headline averages have no reported uncertainty. Evidence for reusable skills—not a self-improving update protocol.

Our read · HarnessRSI

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Reports: held-out tests, transfer, ablations, single-round token costs; partial budget matching.

Does not report: seed variance, leakage audit, baseline tuning budgets or a default-selection split.